Laravel COMPLETE STUDENT CRUD - STEP BY STEP

Step 1: Run These Artisan Commands

Open **Terminal/CMD** in your project folder (school-management-system)

bash

 # 1. Create Model with Migration

```
php artisan make:model Student -m
```

 # 2. Create StudentProfile Model with Migration

```
php artisan make:model StudentProfile -m
```

 # 3. Create Controller (resource controller - gives all CRUD methods)

```
php artisan make:controller StudentController --resource
```

 # 4. Create Factory (for testing data - optional but helpful)

```
php artisan make:factory StudentFactory --model=Student
```

 # 5. Create Seeder

```
php artisan make:seeder StudentSeeder
```

Step 2: After Commands, Replace Code in Each File

Migration 1: Students Table

File: database/migrations/xxxx_xx_xx_xxxxxx_create_students_table.php

Replace the up() method content:

php

```
public function up(): void
{
```

```
    Schema::create('students', function (Blueprint $table) {
        $table->id();
        $table->string('name');
        $table->string('email')->unique();
        $table->string('phone')->nullable();
        $table->string('class')->nullable();
        $table->date('dob')->nullable();
        $table->enum('gender', ['male', 'female', 'other'])->nullable();
        $table->timestamps();
    });
}
```

Migration 2: Student Profiles Table

File: database/migrations/xxxx_xx_xx_xxxxxx_create_student_profiles_table.php

Replace **full file** content:

```
php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    public function up(): void
    {
```

```
        Schema::create('student_profiles', function (Blueprint $table) {
            $table->id();
            $table->foreignId('student_id')->constrained()->onDelete('cascade');
            $table->text('address')->nullable();
            $table->string('father_name')->nullable();
            $table->string('mother_name')->nullable();
            $table->string('emergency_contact')->nullable();
            $table->string('blood_group')->nullable();
            $table->string('profile_picture')->nullable();
            $table->timestamps();
        });
    }

    public function down(): void
    {
        Schema::dropIfExists('student_profiles');
    }
};
```

Model: Student.php

File: app/Models/Student.php

Replace **full file** content:

```
php
<?php

namespace App\Models;
```

```php
use Illuminate\Database\Eloquent\Model;
use Illuminate\Database\Eloquent\Relations\HasOne;
use Illuminate\Database\Eloquent\Relations\BelongsToMany;

class Student extends Model
{
    protected $fillable = [
        'name', 'email', 'phone', 'class', 'dob', 'gender'
    ];

    // One-to-One: Student has one profile
    public function profile(): HasOne
    {
        return $this->hasOne(StudentProfile::class);
    }

    // Many-to-Many: Student belongs to many courses (will use later)
    public function courses(): BelongsToMany
    {
        return $this->belongsToMany(Course::class, 'course_student');
    }
}
```

Model: StudentProfile.php

File: app/Models/StudentProfile.php

Replace **full file** content:

php

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;
use Illuminate\Database\Eloquent\Relations\BelongsTo;

class StudentProfile extends Model
{
    protected $fillable = [
        'student_id', 'address', 'father_name', 'mother_name',
        'emergency_contact', 'blood_group', 'profile_picture'
    ];

    // Inverse of One-to-One
    public function student(): BelongsTo
    {
        return $this->belongsTo(Student::class);
    }
}
```

Controller: StudentController.php

File: app/Http/Controllers/StudentController.php

Replace **full file** content:

```
php
<?php

namespace App\Http\Controllers;

use App\Models\Student;
use App\Models\StudentProfile;
use Illuminate\Http\Request;

class StudentController extends Controller
{
    // Show all students (READ)
    public function index()
    {
        $students = Student::with('profile')->get();
        return view('students.index', compact('students'));
    }

    // Show form to create new student
    public function create()
    {
        return view('students.create');
    }

    // Save new student (CREATE)
    public function store(Request $request)
```

```
{
    $validated = $request->validate([
        'name' => 'required|string|max:255',
        'email' => 'required|email|unique:students',
        'phone' => 'nullable|string',
        'class' => 'nullable|string',
        'dob' => 'nullable|date',
        'gender' => 'nullable|in:male,female,other',
        'address' => 'nullable|string',
        'father_name' => 'nullable|string',
        'mother_name' => 'nullable|string',
        'emergency_contact' => 'nullable|string',
        'blood_group' => 'nullable|string',
    ]);

    // Create Student
    $student = Student::create([
        'name' => $validated['name'],
        'email' => $validated['email'],
        'phone' => $validated['phone'] ?? null,
        'class' => $validated['class'] ?? null,
        'dob' => $validated['dob'] ?? null,
        'gender' => $validated['gender'] ?? null,
    ]);

    // Create Profile
```

```
    StudentProfile::create([
        'student_id' => $student->id,
        'address' => $validated['address'] ?? null,
        'father_name' => $validated['father_name'] ?? null,
        'mother_name' => $validated['mother_name'] ?? null,
        'emergency_contact' => $validated['emergency_contact'] ?? null,
        'blood_group' => $validated['blood_group'] ?? null,
    ]);

    return redirect()->route('students.index')
        ->with('success', 'Student created successfully!');
}

// Show single student details
public function show(Student $student)
{
    $student->load('profile');
    return view('students.show', compact('student'));
}

// Show edit form
public function edit(Student $student)
{
    $student->load('profile');
    return view('students.edit', compact('student'));
}
```

```
// Update student (UPDATE)
public function update(Request $request, Student $student)
{
    $validated = $request->validate([
        'name' => 'required|string|max:255',
        'email' => 'required|email|unique:students,email,' . $student->id,
        'phone' => 'nullable|string',
        'class' => 'nullable|string',
        'dob' => 'nullable|date',
        'gender' => 'nullable|in:male,female,other',
        'address' => 'nullable|string',
        'father_name' => 'nullable|string',
        'mother_name' => 'nullable|string',
        'emergency_contact' => 'nullable|string',
        'blood_group' => 'nullable|string',
    ]);

    // Update Student
    $student->update([
        'name' => $validated['name'],
        'email' => $validated['email'],
        'phone' => $validated['phone'] ?? null,
        'class' => $validated['class'] ?? null,
        'dob' => $validated['dob'] ?? null,
        'gender' => $validated['gender'] ?? null,
```

```
    ]);

    // Update or Create Profile
    if ($student->profile) {
        $student->profile->update([
            'address' => $validated['address'] ?? null,
            'father_name' => $validated['father_name'] ?? null,
            'mother_name' => $validated['mother_name'] ?? null,
            'emergency_contact' => $validated['emergency_contact'] ?? null,
            'blood_group' => $validated['blood_group'] ?? null,
        ]);
    } else {
        StudentProfile::create([
            'student_id' => $student->id,
            'address' => $validated['address'] ?? null,
            'father_name' => $validated['father_name'] ?? null,
            'mother_name' => $validated['mother_name'] ?? null,
            'emergency_contact' => $validated['emergency_contact'] ?? null,
            'blood_group' => $validated['blood_group'] ?? null,
        ]);
    }

    return redirect()->route('students.index')
        ->with('success', 'Student updated successfully!');
}
```

```
    // Delete student (DELETE)
    public function destroy(Student $student)
    {
        $student->delete();
        return redirect()->route('students.index')
            ->with('success', 'Student deleted successfully!');
    }
}
```

Routes: web.php

File: routes/web.php

Add these lines:

php

```
<?php

use App\Http\Controllers\StudentController;

// Student CRUD Routes (all 7 methods)
Route::resource('students', StudentController::class);

// OR you can write individually:
// Route::get('/students', [StudentController::class, 'index'])->name('students.index');
// Route::get('/students/create', [StudentController::class, 'create'])->name('students.create');
// Route::post('/students', [StudentController::class, 'store'])->name('students.store');
// Route::get('/students/{student}', [StudentController::class, 'show'])->name('students.show');
// Route::get('/students/{student}/edit', [StudentController::class, 'edit'])->name('students.edit');
```

```
// Route::put('/students/{student}', [StudentController::class, 'update'])->name('students.update');

// Route::delete('/students/{student}', [StudentController::class, 'destroy'])->name('students.destroy');


// Home route (your existing)

Route::get('/', function () {

    return view('welcome');

});
```

Step 3: Create Views Folder & Files

- **Create folder structure:**

text

resources/views/students/

- **Create 4 files inside students/ folder:**

File 1: index.blade.php

blade

```
@extends('layouts.app')


@section('title', 'Students')

@section('page-title', 'Students Management')


@section('content')


<div style="display: flex; justify-content: space-between; margin-bottom: 20px;">

    <h3>Students List</h3>
```

```
    <a href="{{ route('students.create') }}" style="background: #2c3e50; color: white; padding: 8px 16px; text-decoration: none;">+ Add Student</a>
</div>

@if(session('success'))
    <div style="background: green; color: white; padding: 10px; margin-bottom: 15px;">{{ session('success') }}</div>
@endif

<table border="1" width="100%" cellpadding="10">
    <tr style="background: #34495e; color: white;">
        <th>ID</th><th>Name</th><th>Email</th><th>Class</th><th>Actions</th>
    </tr>
    @foreach($students as $student)
    <tr>
        <td>{{ $student->id }}</td>
        <td>{{ $student->name }}</td>
        <td>{{ $student->email }}</td>
        <td>{{ $student->class ?? '-' }}</td>
        <td>
            <a href="{{ route('students.show', $student->id) }}">View</a>
            <a href="{{ route('students.edit', $student->id) }}">Edit</a>
            <form action="{{ route('students.destroy', $student->id) }}" method="POST" style="display:inline;">
                @csrf @method('DELETE')
                <button type="submit" onclick="return confirm('Delete?')">Delete</button>
            </form>
```

```
        </td>
    </tr>
    @endforeach
</table>

@endsection
```

File 2: create.blade.php

blade

```
@extends('layouts.app')

@section('title', 'Add Student')
@section('page-title', 'Add New Student')

@section('content')

<form action="{{ route('students.store') }}" method="POST">
    @csrf

    <label>Name:</label>
    <input type="text" name="name" required><br><br>

    <label>Email:</label>
    <input type="email" name="email" required><br><br>

    <label>Phone:</label>
    <input type="text" name="phone"><br><br>
```

```
    <label>Class:</label>
    <input type="text" name="class"><br><br>

    <label>Father Name:</label>
    <input type="text" name="father_name"><br><br>

    <label>Address:</label>
    <textarea name="address"></textarea><br><br>

    <button type="submit">Save Student</button>
    <a href="{{ route('students.index') }}">Cancel</a>
</form>

@endsection
```

File 3: edit.blade.php

blade

```
@extends('layouts.app')

@section('title', 'Edit Student')
@section('page-title', 'Edit Student: ' . $student->name)

@section('content')

<form action="{{ route('students.update', $student->id) }}" method="POST">
    @csrf
```

```
    @method('PUT')

    <label>Name:</label>
    <input type="text" name="name" value="{{ $student->name }}" required><br><br>

    <label>Email:</label>
    <input type="email" name="email" value="{{ $student->email }}" required><br><br>

    <label>Phone:</label>
    <input type="text" name="phone" value="{{ $student->phone }}"><br><br>

    <label>Class:</label>
    <input type="text" name="class" value="{{ $student->class }}"><br><br>

    <label>Father Name:</label>
    <input type="text" name="father_name" value="{{ $student->profile->father_name ?? ''
}}"><br><br>

    <label>Address:</label>
    <textarea name="address">{{ $student->profile->address ?? '' }}</textarea><br><br>

    <button type="submit">Update Student</button>
    <a href="{{ route('students.index') }}">Cancel</a>
</form>

@endsection
```

File 4: show.blade.php

blade

```
@extends('layouts.app')

@section('title', 'Student Details')
@section('page-title', 'Student Details')

@section('content')

<p><strong>Name:</strong> {{ $student->name }}</p>
<p><strong>Email:</strong> {{ $student->email }}</p>
<p><strong>Phone:</strong> {{ $student->phone ?? '-' }}</p>
<p><strong>Class:</strong> {{ $student->class ?? '-' }}</p>
<p><strong>Father Name:</strong> {{ $student->profile->father_name ?? '-' }}</p>
<p><strong>Address:</strong> {{ $student->profile->address ?? '-' }}</p>

<a href="{{ route('students.edit', $student->id) }}">Edit</a>
<a href="{{ route('students.index') }}">Back</a>

@endsection
```

Step 4: Run These Final Commands

bash

```
# 1. Run migrations (create tables in database)
php artisan migrate
```

 # 2. (Optional) Clear cache if any error

```
php artisan config:clear

php artisan cache:clear
```

 # 3. Start the server

```
php artisan serve
```

Step 5: Test in Browser

1. Open: **http://127.0.0.1:8000/students**